

Revised Load Interconnection System Impact Study Report

For Queue Project Q13-0220

Malaspina 138 kV
April 20, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0220. The Interconnection Customer proposes a 91.5 MW large load (data center) facility interconnecting at the Malaspina 138 kV substation in the North Carolina area. The request was received on March 30, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 128 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 1 overloaded facility. The total estimated cost of the network upgrades allocated to this project is \$1,374,000.

3. Network Impacts

3.1 Load Deliverability Analysis

1. Facility: MALASPINA to OLS ENERGY CHINO 138 kV line
Contingency: MECKLENBURG POWER STATION to PERSON 345 kV
Loading: 102.1% (Rating: 384 MVA)
DFAX: 81.8%
Pre-project loading: 82.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MALASPINA to OLS ENERGY CHINO 138 kV line	Reconductor line	\$4,551,000	30% (\$1,374,000)
	Total allocated to project		\$1,374,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.